ANALYTICS FOR BUSINESS LAB

acceleration
A WPP Company

GROUP M - MARKETING MIX MODELLING

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INTRODUCTION AND OBJECTIVES

- Understanding the dynamics of business and what drives sales.
- Understanding the effects of each factor(promo, media, etc) on sales and how they change on a year-on-year basis.
- Deep dive into media investments and contributions.
- Total effectiveness, total Return on Investments and the optimizing the media mix.

The background provided for this MMM project:

- A comparatively young beauty brand that has gained prominence in the market over the last five years.
- Their primary goal is to increase their market share, by increasing the recognition of their product in the market. To achieve this, they want to optimize their media and promotional investments.

METHODOLOGY

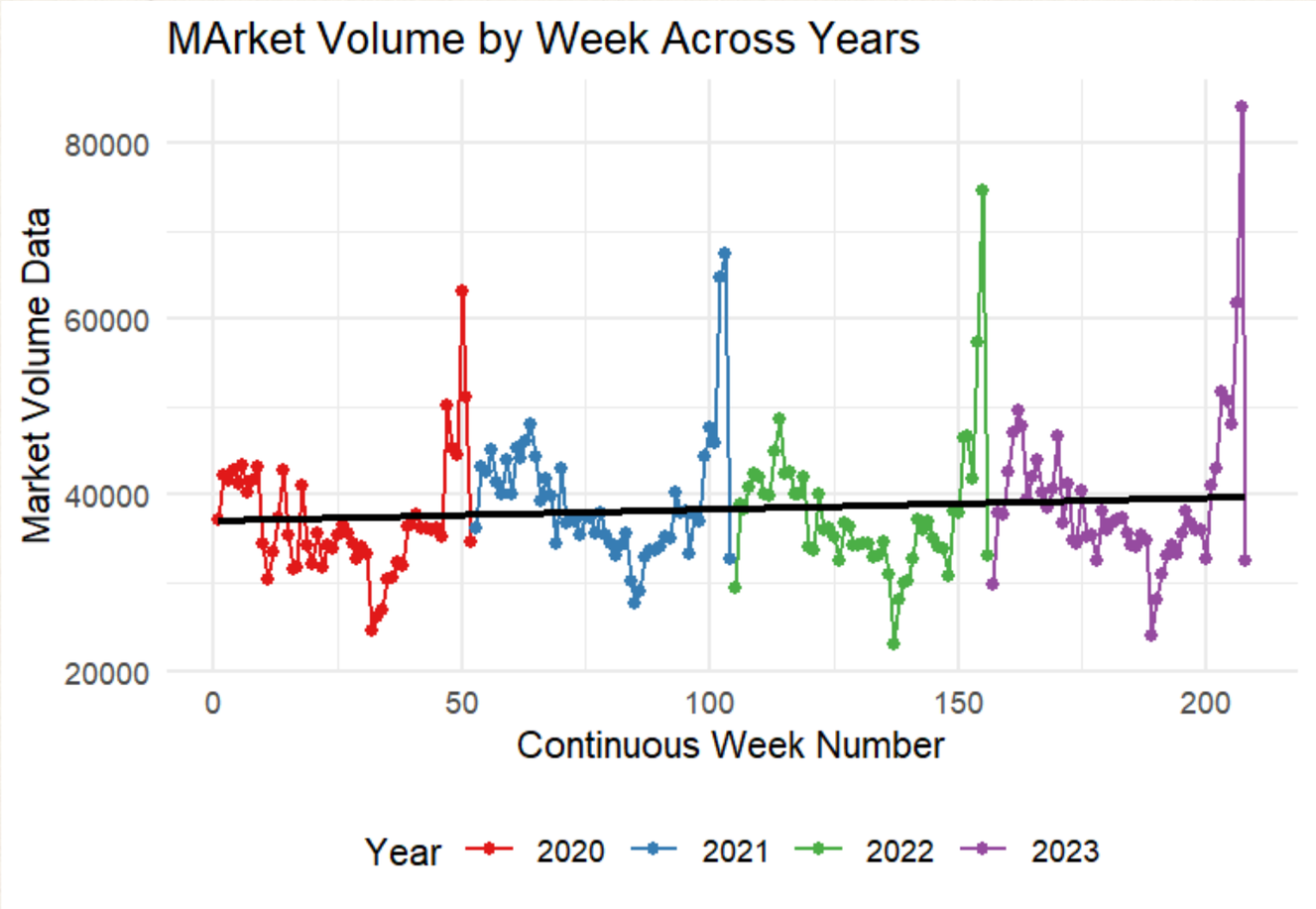
To develop a Mix Marketing Model for this brand, we chose to use Robyn, which is a Semi-Automated Marketing Mix Modeling (MMM) aiming to reduce human bias by means of ridge regression and evolutionary algorithms. We have provided a detailed analysis of how we adopted it for this project in the ‘Project Methodology’ slide.

DATA INSIGHTS

UNDERSTANDING MARKET DYNAMICS

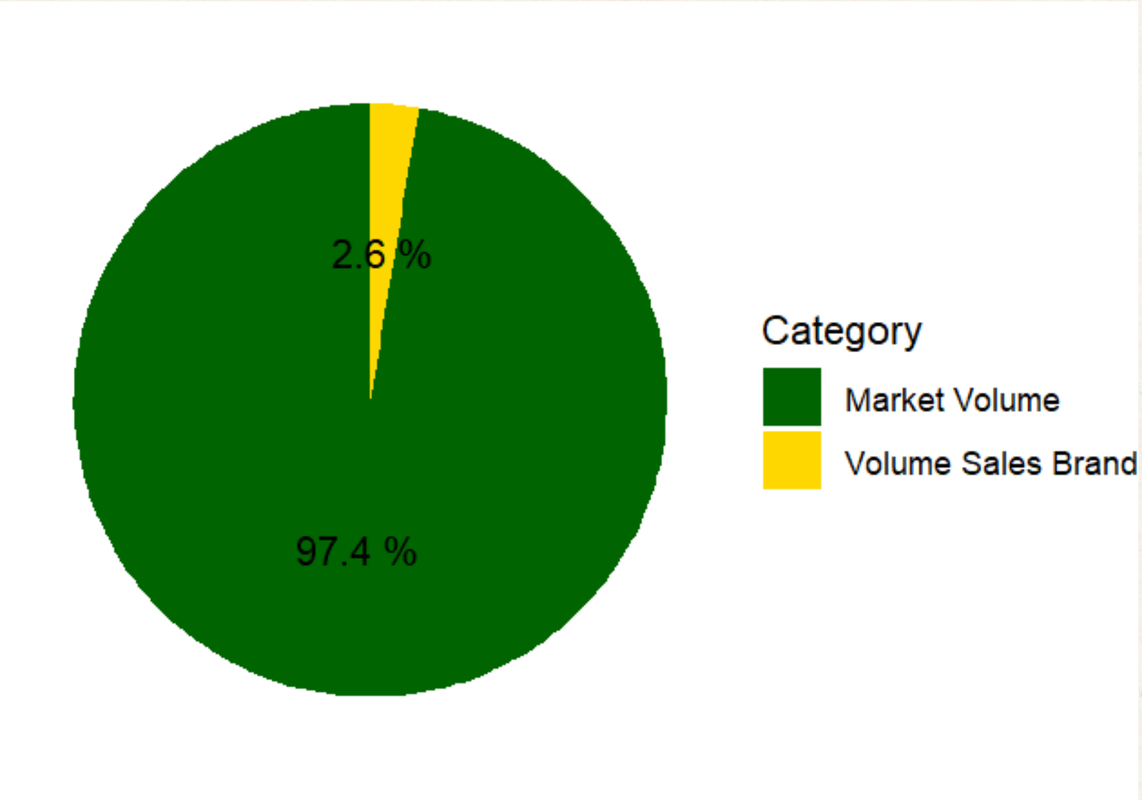
- We checked the trend of the market and the brand in terms of the amount of volume that was being sold.
- Based on this graph we can understand that the volume sales in this market are not growing, but they are not declining either. The trend also shows that the product sells a lot in during certain times a year, indicating strong seasonality of sales.

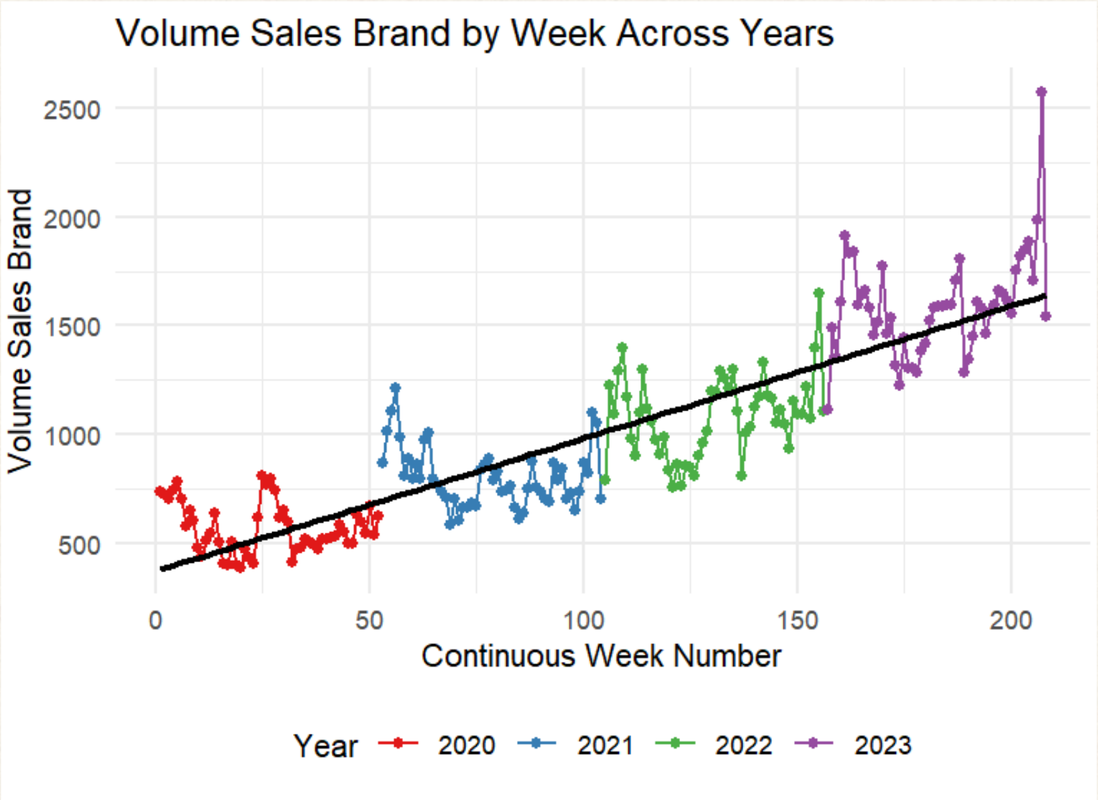
- For our brand, we can clearly see that its volume sales are growing constantly through the years. This means that the brand is performing really well and is getting recognized by a larger group of people. It's current market share of 2.6% also shows that it has a great oppurtunity to expand.



DATA INSIGHTS

UNDERSTANDING MARKET DYNAMICS





Based on these comparisons, we concluded that the current MMM structure present, works, but we would have to optimise it, to increase the return on investments (ROI) and the return on ad spend(ROAS).

DATA INSIGHTS

In order to better understand how the investments and impressions are correlated and affecting our dependent variable we used correlation matrices.

INVESTMENTS TV	V1
INVESTMENTS PRINT	V2
INV YOUTUBE	V3
INV OTHER VIDEO PUBLISHERS	V4
INV SOCIAL META	V5
INV GOOGLE SEARCH	V6
INV AMAZON SEARCH	V7
INVESTMENT TIKTOK	V8
INVESTMENT TWITCH	V9
INVESTMENT INFLUENCER MARKETING TIKTOK	V10
INVESTMENT INFLUENCER MARKETING INSTAGRAM	V11

	Value Sales Brand	*V1*	*V2*	*V3*	*V4*	*V5*	*V6*	*V7*	*V8*	*V9*	*V10*	*V11*
Value Sales Brand	1											
V1	0.176	1										
V2	0.088	0.157	1									
V3	0.556	0.223	0.067	1								
V4	0.200	0.018	-0.026	0.417	1							
V5	0.531	0.178	0.020	0.759	0.323	1						
V6	0.827	0.114	0.062	0.440	0.320	0.453	1					
V7	0.625	0.062	0.047	0.445	0.085	0.319	0.564	1				
V8	0.248	0.003	-0.013	0.354	0.118	0.380	0.197	0.273	1			
V9	0.407	0.111	-0.020	0.402	0.360	0.407	0.407	0.130	0.122	1		
V10	0.301	0.008	-0.021	0.184	-0.023	0.162	0.269	0.197	0.032	0.240	1	
V11	0.243	0.100	-0.028	0.120	0.144	0.200	0.290	0.152	0.067	-0.013	0.290	1

GRP TV	V12
IMPRESSIONS YOUTUBE	V13
IMPRESSIONS OTHER VIDEO PUBLISHERS	V14
IMPRESSIONS SOCIAL META	V15
IMPRESSIONS SPOTIFY	V16
IMPR. TIKTOK	V17
IMPR. TWITCH	V18
IMPRESSIONS INFLUENCER MARKETING TIKTOK	V19
IMPRESSIONS INFLUENCER MARKETING INSTAGRAM	V20

	Value Sales Brand	**V12**	**V13**	**V14**	**V15**	**V16**	**V17**	**V18**	**V19**	**V20**
Value Sales Brand	1.000									
V12	0.155	1.000								
V13	0.589	0.138	1.000							
V14	0.229	0.015	0.387	1.000						
V15	0.397	0.201	0.721	0.296	1.000					
V16	0.176	-0.044	0.285	-0.070	0.121	1.000				
V17	0.394	-0.011	0.505	0.156	0.503	0.328	1.000			
V18	0.406	0.043	0.392	0.368	0.307	-0.044	0.261	1.000		
V19	0.269	0.030	0.156	0.031	0.116	0.019	0.078	0.241	1.000	
V20	0.125	0.114	0.085	0.145	0.122	0.076	0.023	0.001	0.182	1.000

DATA INSIGHTS

In the second correlation matrix, between the Impressions and the value sales brand, we can observe that impressions in Google Search, Social Meta, and Amazon Search have high correlation with each others. We could infer that when our target customers discover this product on Social Meta, they go to Google or Amazon to know more about it. These inferences align with the correlation matrix of Investments and Value Sales Brand.

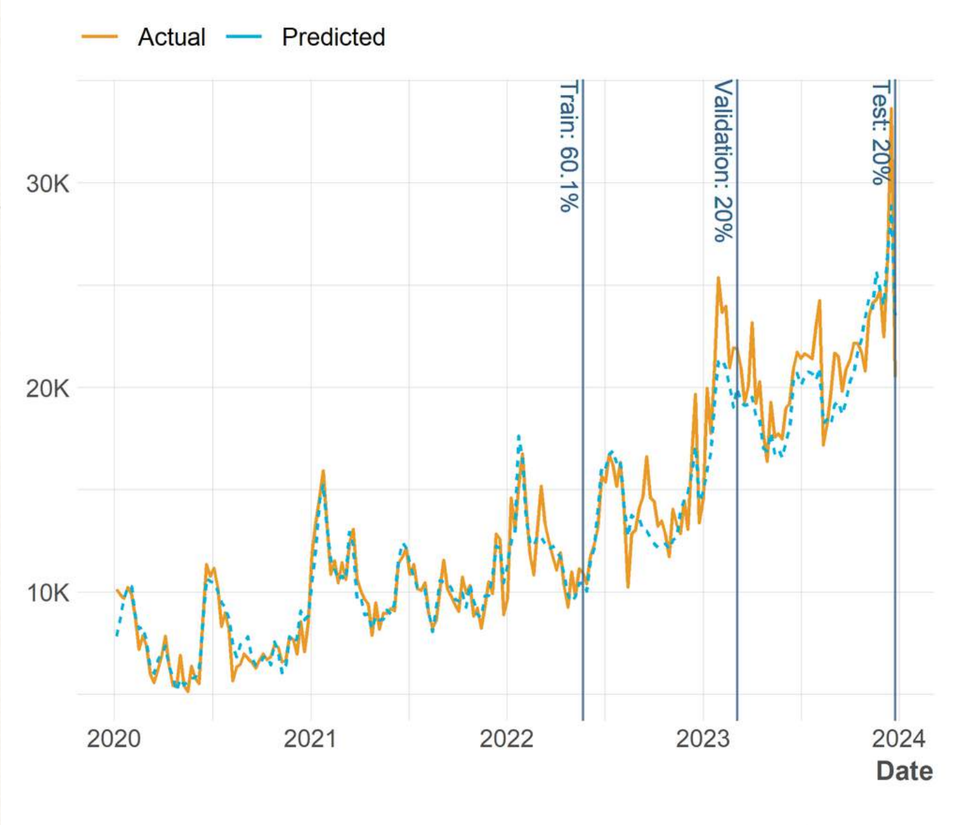
After deciding on factors like the dependent and independent variables we created our MMM model using Robyn, which gave us the following results.

To judge the efficiency of our model, we ran statistical tests and for the model used in this analysis, we got the following results:

- Adjusted R-square = 0.8878
- Root Mean Squared Error = 0.0645

Based on these results we selected our final MMM model.

The graph (Actual vs Predicted) on the right also shows how much our model fits the actual data.

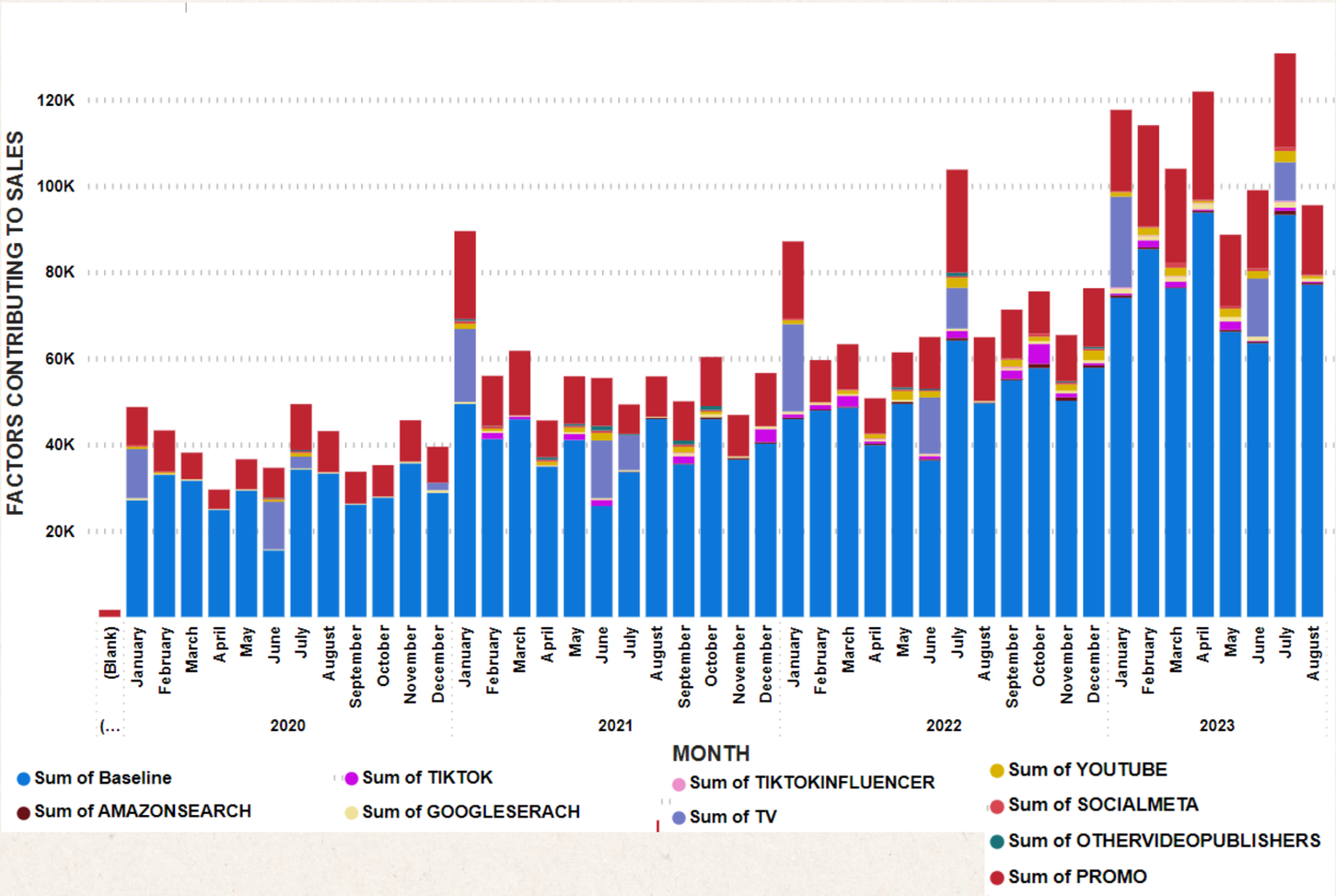


DATA INSIGHTS

The graph on the right is the Sales Decomposition Graph. This graph differentiates the base sales from the incremental factors that are contributing to the overall sales.

Inferences:

- We can see that most of the sales are comprised of base sales, which indicates a strong influence of trend. This means that, even in the absence of media and promotional activities, our product would still sell in the market because there is a strong demand for the product.
- The next strong presences is of promotional activities. We can see that the sales are higher during the months of January each year and the the share of promo is also higher particularly in those months compared to the rest. This means that our promos are the most effective during the holidays.

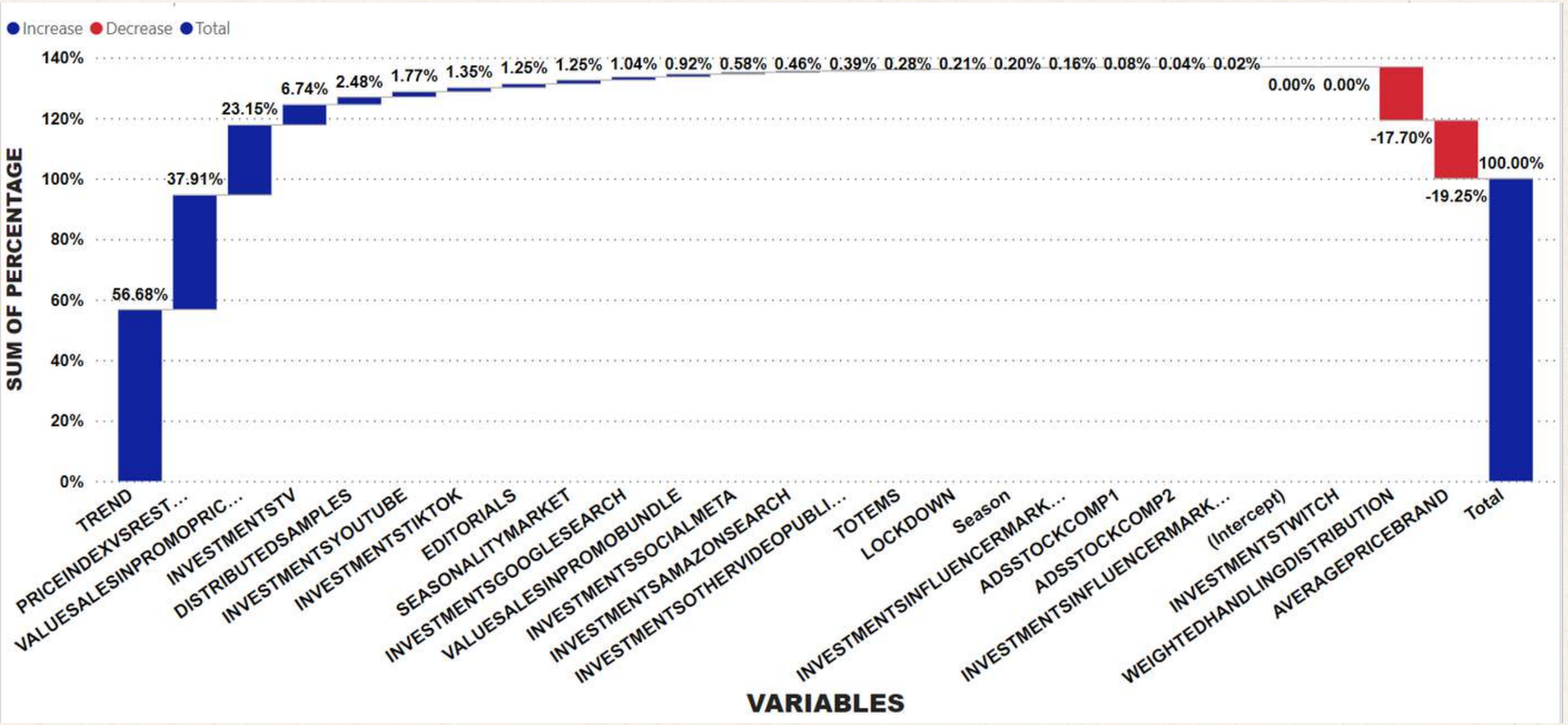


To understand the effect of each variable on the sales we created a Response Decomposition chart based on our robyn model. This chart helps us in understanding how much each variable is contributing towards the final sale.

DATA INSIGHTS

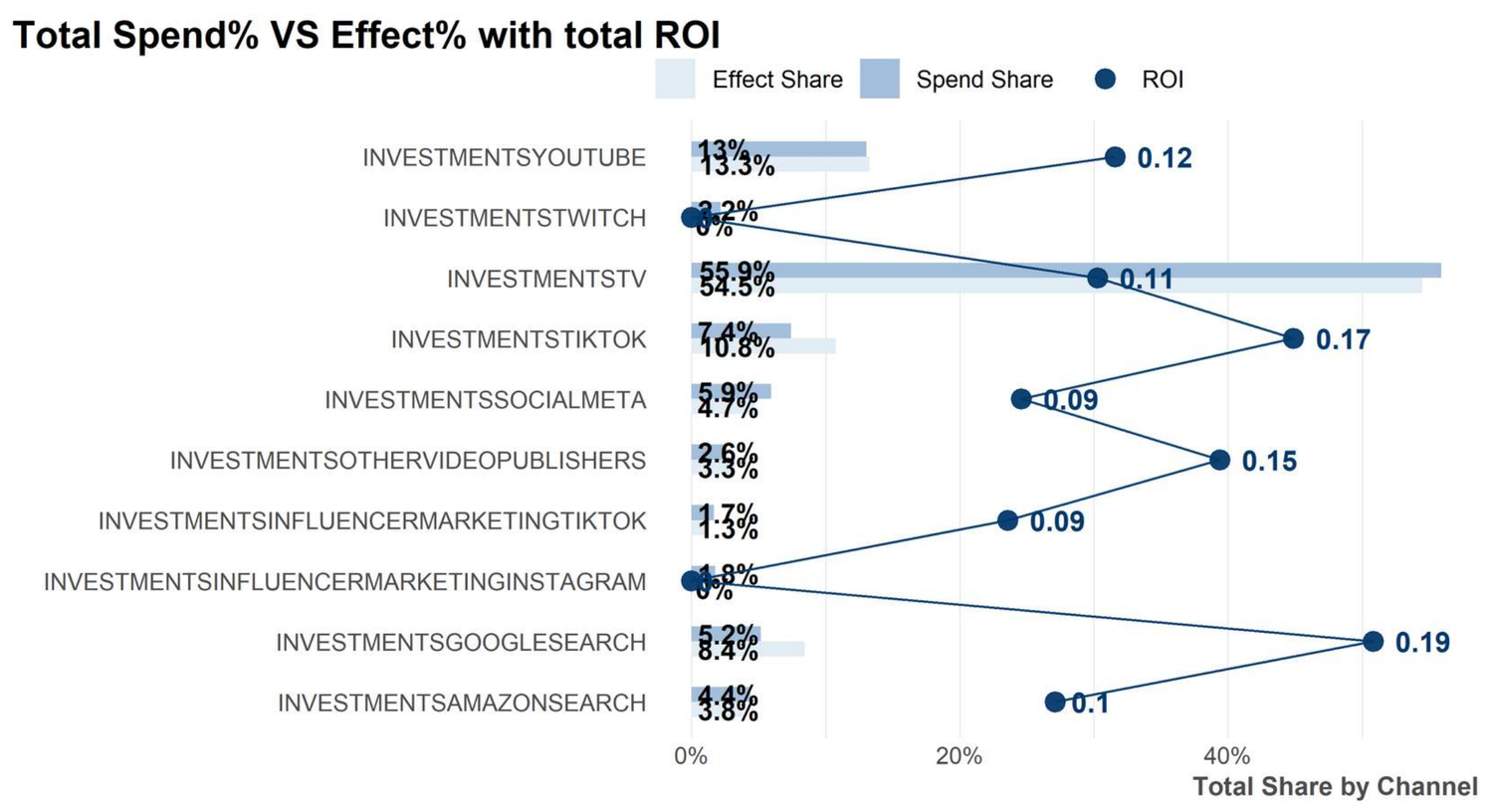
CONTRIBUTION OF EACH VARIABLE IN GENERATING SALES:

Based on this decomposition chart we can see that the majority of the sales can be attributed to the trend and promotional activities. The total contribution of the media channels are around 15%. In order to optimise the contribution of the media channels we first look at the Return on Investments(ROI) of each channel. Then we can decide on which channel we can invest the most to get better ROIs.



MARKETING MIX EFFECTIVENESS

Based on our model, the Return on Investment for each media channels are:

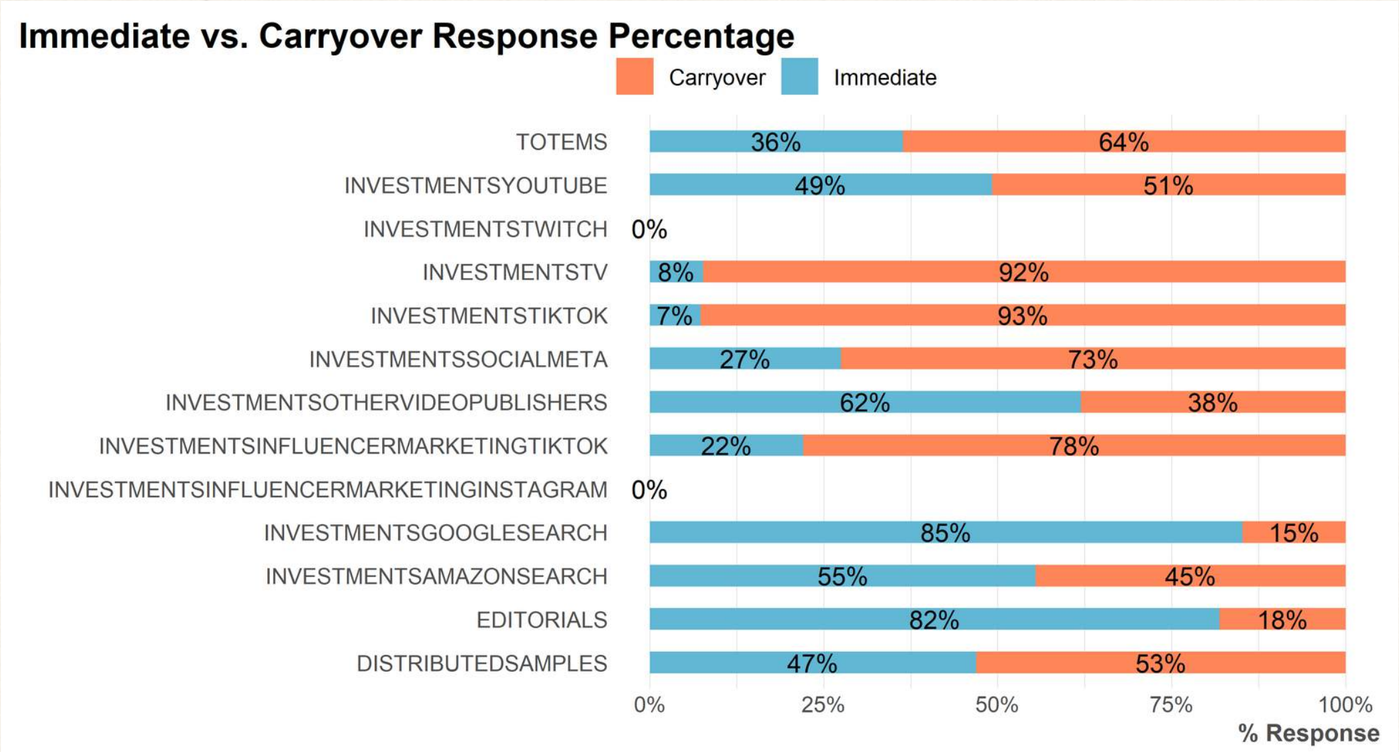


GOOGLE SEARCH	0.192909911
TIKTOK	0.170407145
OTHER VIDEO PUBLISHERS	0.149494784
YOUTUBE	0.119850513
TV	0.114893473
AMAZON SEARCH	0.102964787
SOCIAL META	0.093266564
INFLUENCER MARKETING TIKTOK	0.089446075
TWITCH	0
INFLUENCER MARKETING INSTAGRAM	0

This table contains the ROI values in decreasing order. We can infer from this data that the most effective channels for this brand are Google search, TikTok, other video publishers, Youtube and investments in TV ads.
Therefore to create an optimum MMM budget we need to divert some of the investments into these channels.

MARKETING MIX EFFECTIVENESS

To optimize our media mix, we also looked at the effect of each media channel.



Inferences:

- Based on the results of the ROI analysis and the carryover effect of media channels, we believe that the best influencer category for our brand would be TikTok. We understand that our target customers are responding better to video-based ad formats. This is in line with the fact that the ROI for YouTube is also comparatively high.
- Since traditional media channels and video-based media channels work better for this brand we also believe that majority of our target customers are not very active on social media platforms like Instagram and Twitch, indicating that this customer group are of an older age category. Hence, focusing our investments on traditional media channels and few of the online channels(Social Meta, TikTok, Google search and Amazon Search) might yield better results in terms of overall sales.

BUDGET ALLOCATION

The optimum budget allocation based on our MMM model are defined as follows:

Budget Allocation per Channel*

Paid Channels	Initial				Bounded				Bounded x3			
	spend%	response%	ROAS	mROAS	spend%	response%	ROAS	mROAS	spend%	response%	ROAS	mROAS
INVESTMENTSAMAZONSEARCH	4.9%	5.8%	0.06	0.06	7.3%	6.7%	0.06	0.06	12.2%	8%	0.06	0.04
INVESTMENTSGOOGLESEARCH	6.2%	15.6%	0.13	0.23	9.3%	23.6%	0.17	0.25	15.5%	34%	0.19	0.17
INVESTMENTSINFLUENCERMARKETINGINSTAGRAM	2.1%	0%	0	0	0%	0%	0	0	0%	0%	0	0
INVESTMENTSINFLUENCERMARKETINGTIKTOK	3%	0.1%	0	0	2.1%	0.1%	0	0	0.3%	0%	0	0
INVESTMENTSOTHERVIDEOPUBLISHERS	2.1%	4.9%	0.12	0.12	3.2%	5.7%	0.12	0.12	5.3%	7.3%	0.12	0.11
INVESTMENTSSOCIALMETA	6.7%	4.2%	0.03	0.04	10%	5.1%	0.03	0.04	0.7%	0.3%	0.04	0.02
INVESTMENTSTIKTOK	6%	7.4%	0.07	0.02	4.2%	5.4%	0.09	0.02	0.6%	3.6%	0.52	0.01
INVESTMENTSTV	49.1%	41.3%	0.04	0.02	42.2%	30.1%	0.05	0.03	29.1%	19.6%	0.06	0.03
INVESTMENTSTWITCH	5.5%	0%	0	0	0%	0%	0	0	0%	0%	0	0
INVESTMENTSYOUTUBE	14.5%	20.6%	0.08	0.07	21.8%	23.3%	0.07	0.06	36.4%	27.1%	0.06	0.04

BUDGET ALLOCATION

Terminologies:

- *INITIAL*: The initial column represents the baseline scenario with the current budget allocation. It shows what is the overall spend% in each channel and what is the resulting response%. Based on these results it gives us the Return on Ad Spend(ROAS) and the Marginal ROAS(mROAS). The mROAS shows the efficiency of the additional spend in generating responses for particular channels.
- *BOUNDED:* In this part, we keep our total budget allocation constant, but we are redistributing it based on the results of the effect share of each channel.
- *BOUNDEDX3*: The boundedx3 represents a more aggressive budget allocation based on the same effect share and keeping the same budget constraints.

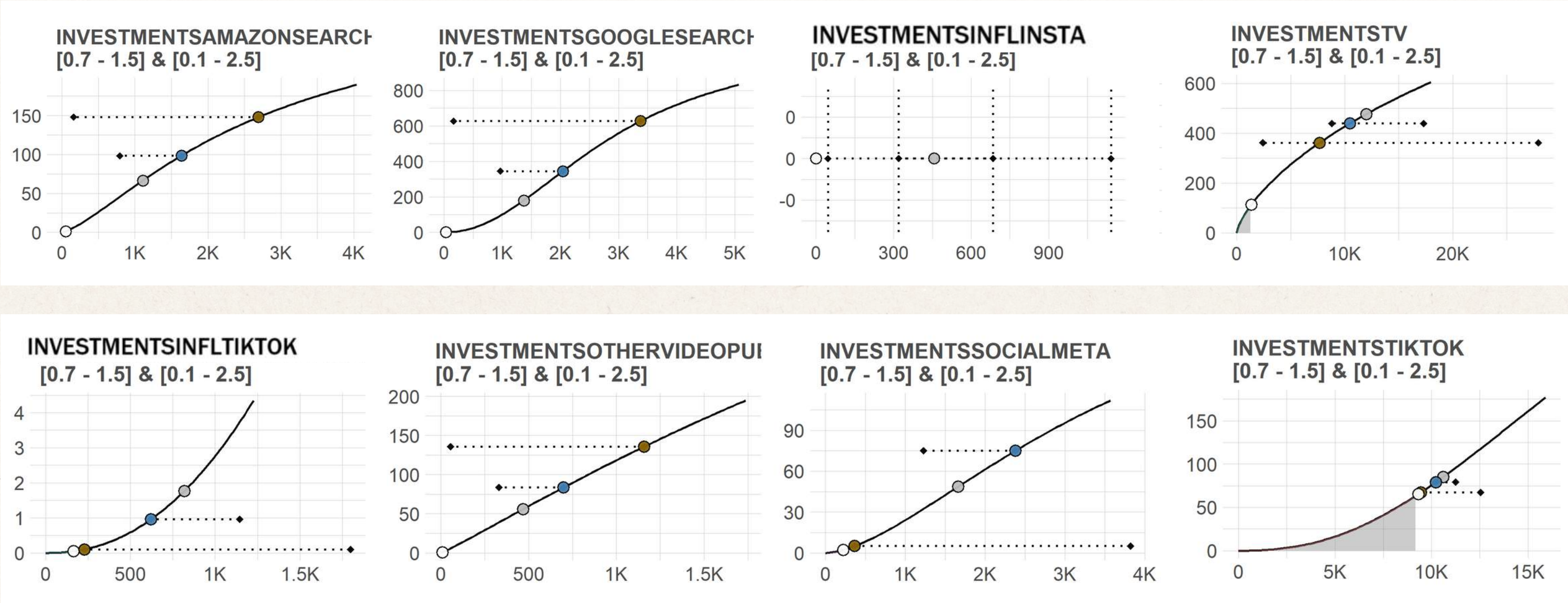
INSIGHTS

- If we take the example of Investments TV, which has an initial spend% of 49.1% and ROAS of 0.04, if we reduce it's spend% and reallocate it to a different channel, we get a better ROAS for TV of 0.05(bounded) and 0.06(boundedx3). This means that the current investments in TV have already reached the latter part of saturation curve, where additional investment can not bring much increase in sales .
- By reallocating the investment from TV, we increase our investment in Amazon and Google search, we can see that we get better responses. From 5.8% to 8%(boundedx3) for Amazon search and from15.6% to 34%(bounded x3) for Google search.
- Since the effects of investments in Tiktok influencers and investments are very low(close to 0%), we should re-consider the influencers or the streamers that we are currently investing with or we just stop our investments.
- This relationship between the Initial, bounded and boundedx3, and the results of ROAS also gives us the idea of how saturated a media channel has become.

BUDGET ALLOCATION

INSIGHTS:

- The saturation curves for each media channel are as follows:



BUDGET ALLOCATION

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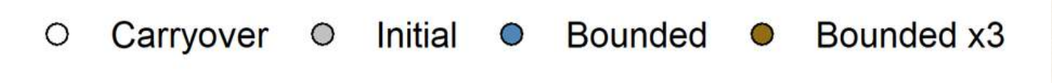
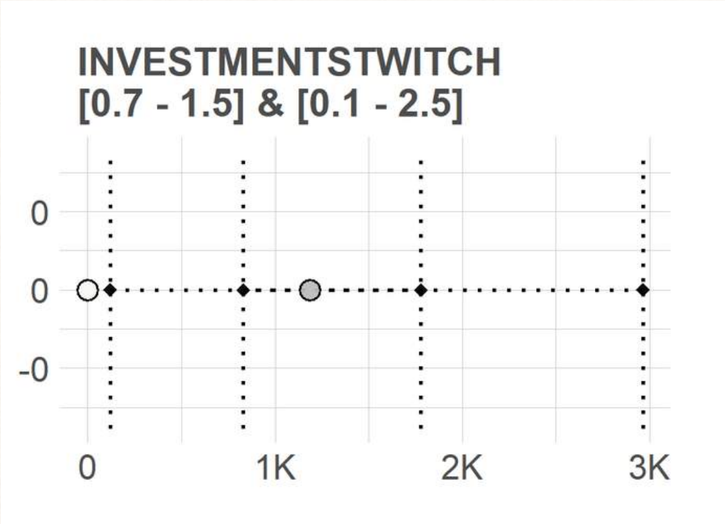
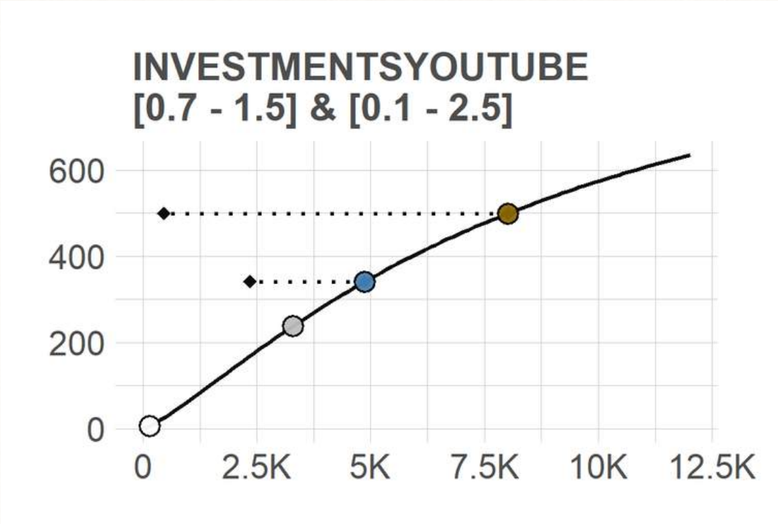
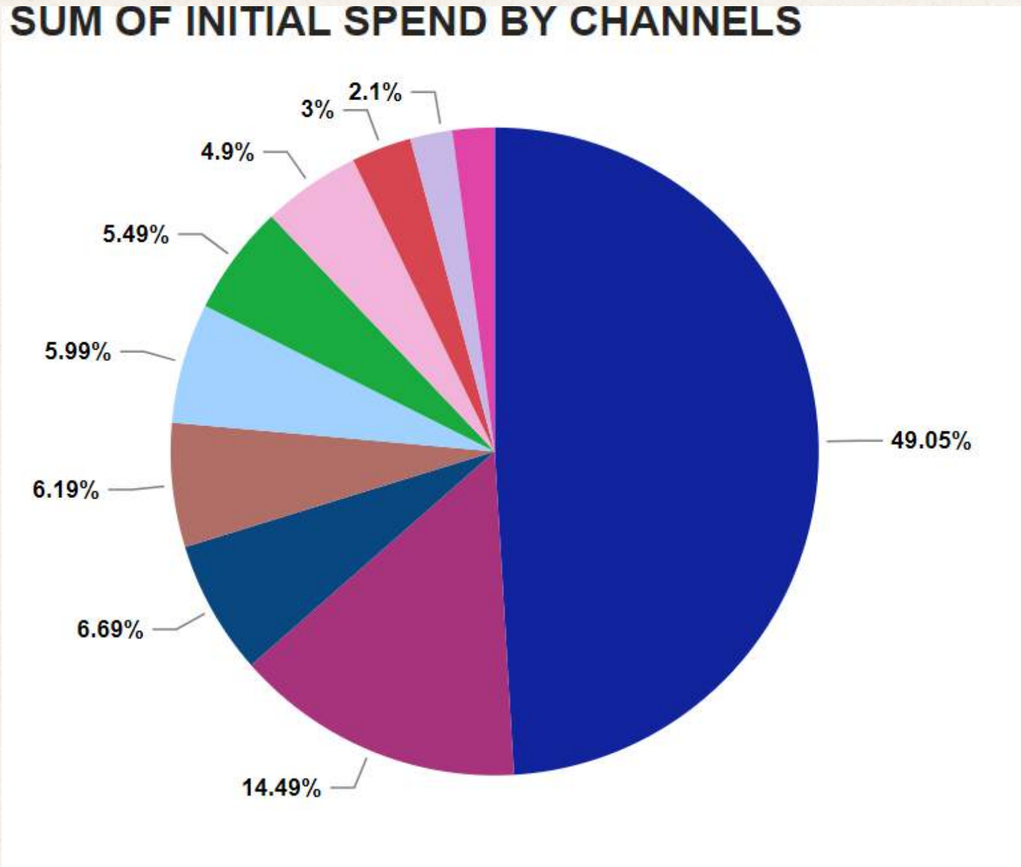
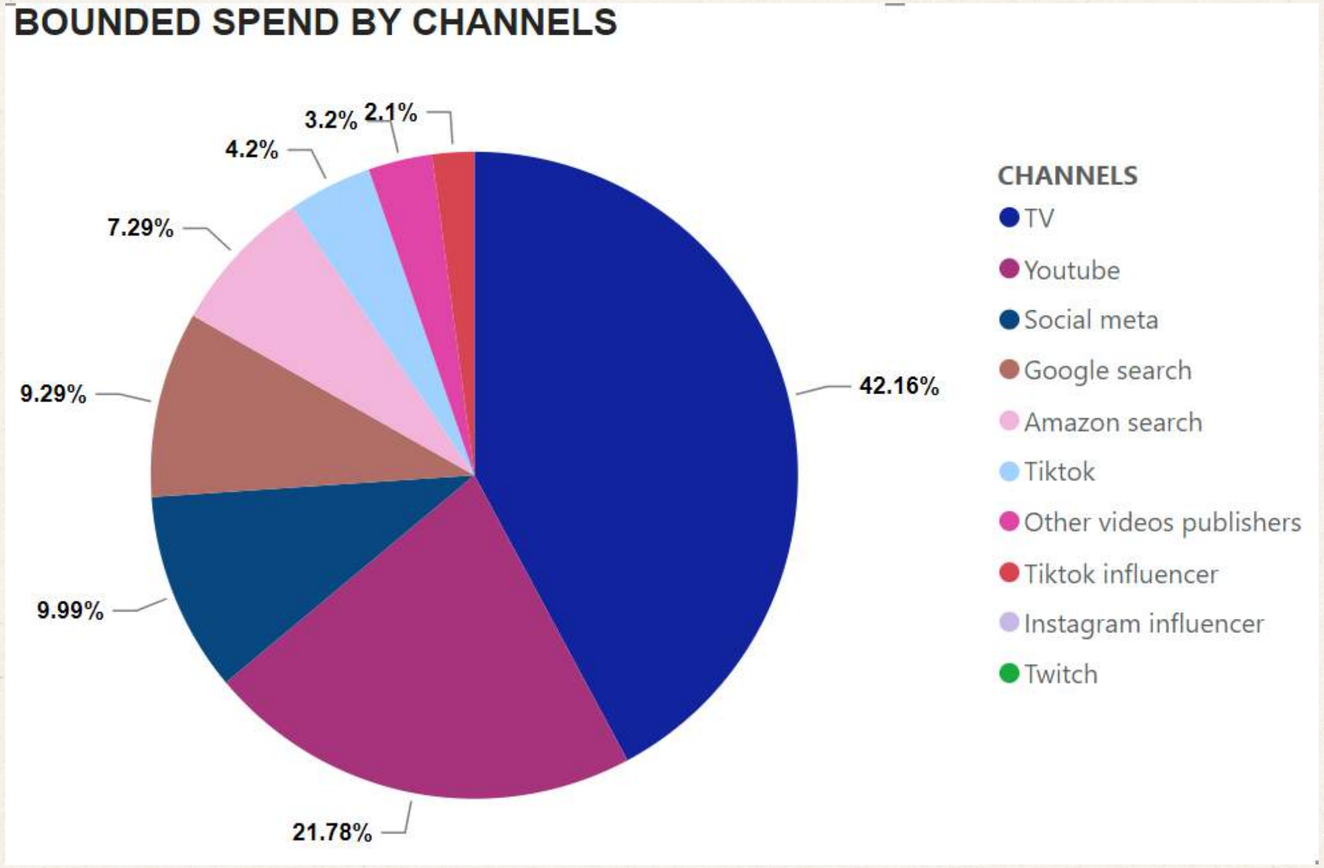






INSIGHTS:

- Based on these saturation curves, we infer that increasing the investments in Youtube, Google Search, Amazon Search, and Other video publishers would yield better responses and higher ROAS.
- The investment in TV has almost reached it’s saturation point and further investments might lead to cannibalization of the media channel.
- For the investments in Social meta, we can see that a further increase of a small magnitude of investment might yield better responses, but if the investments are increased beyond this threshold, it might lead to cannibalization.

BUDGET ALLOCATION

Based on all the inferences and results from our robyn model, we created the most optimized budget based on the current total investment.